

FERGANA, Uzbekistan

WMO: 386180

Lat: 40.367N

Lon: 71.750E

Elev: 1980

StdP: 13.67

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.4	20.9	11.0	10.4	19.0	15.4	12.9	24.3	12.9	39.3	9.7	36.0	2.6	120	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	98.8	69.2	96.6	68.9	94.5	68.4	73.2	92.3	71.8	91.1	70.5	89.4	4.8	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	108.0	82.8	65.5	101.2	81.7	63.8	95.5	80.9	38.3	92.8	36.9	91.5	35.8	89.6	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.2	11.5	9.0	DB	13.5	103.4	6.9	2.0	8.6	104.9	4.5	106.1	0.6	107.2	-4.4	108.7
			WB	12.7	76.8	6.9	1.6	7.7	77.9	3.7	78.8	-0.1	79.7	-5.1	80.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	33.6	37.9	50.6	61.6	70.8	78.8	82.4	80.0	71.4	59.1	46.3	36.0	(6)
	DBStd	18.20	6.33	7.53	7.59	7.11	5.63	4.63	3.86	4.27	5.34	6.76	7.12	5.72	(7)
	HDD50	1538	509	340	86	7	0	0	0	0	0	11	152	433	(8)
	HDD65	4010	974	757	447	142	16	0	0	0	12	202	562	898	(9)
	CDD50	4883	0	3	105	357	646	863	1004	930	641	294	40	0	(10)
	CDD65	1878	0	0	1	41	197	413	539	465	203	19	0	0	(11)
	CDH74	22026	0	0	10	378	1953	4953	6983	5479	2069	200	1	0	(12)
	CDH80	11047	0	0	0	87	720	2551	3903	2943	811	32	0	0	(13)
(14) Wind	WSAvg	3.4	2.5	2.8	3.5	3.9	4.1	4.3	4.0	3.8	3.5	3.1	2.8	2.7	(14)
(15) Precipitation	PrecAvg	6.9	0.7	0.9	1.0	0.7	0.8	0.4	0.2	0.1	0.2	0.7	0.6	0.6	(15)
	PrecMax	13.5	2.8	3.7	2.7	2.8	3.4	1.3	1.0	0.4	0.9	5.3	3.3	2.0	(16)
	PrecMin	3.2	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.4	0.6	0.8	0.6	0.6	0.6	0.4	0.2	0.1	0.2	1.0	0.7	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	61.8	75.4	86.7	93.5	100.5	102.6	100.3	93.4	84.3	71.2	54.4	(19)
		MCWB	44.0	48.3	56.1	63.8	66.3	67.9	69.8	70.8	68.2	63.6	57.2	45.3	(20)
	2%	DB	48.4	55.5	70.7	82.4	90.3	97.1	100.0	97.1	90.5	79.1	65.5	50.2	(21)
		MCWB	41.7	46.1	54.4	62.1	65.5	67.8	69.2	69.9	66.4	61.0	53.4	43.1	(22)
	5%	DB	45.0	51.8	67.1	78.4	87.4	94.7	97.2	95.1	87.9	75.4	60.8	47.3	(23)
		MCWB	39.9	44.2	53.4	60.4	64.3	67.4	68.9	69.3	65.2	58.7	51.2	41.4	(24)
	10%	DB	42.4	48.6	63.3	74.7	84.1	91.8	95.0	93.0	85.0	71.8	57.1	44.6	(25)
		MCWB	38.4	42.5	51.9	59.1	63.4	66.6	68.6	68.5	63.5	57.1	49.0	39.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	49.6	58.3	66.8	70.9	73.8	75.4	75.0	71.1	65.0	58.8	47.2	(27)
		MCDB	50.7	57.6	70.1	81.3	87.2	92.6	94.6	93.3	89.2	80.2	68.9	52.1	(28)
	2%	WB	42.7	47.2	56.0	64.4	68.0	71.5	73.3	72.9	68.8	62.2	54.5	44.1	(29)
		MCDB	47.4	54.1	67.8	78.1	85.1	90.8	93.0	91.7	86.9	76.4	62.6	48.8	(30)
	5%	WB	40.5	45.2	54.3	62.3	66.4	70.1	71.9	71.5	66.8	60.3	52.1	42.1	(31)
		MCDB	44.0	50.8	64.7	74.4	82.5	89.3	91.5	90.6	84.0	72.7	59.6	46.1	(32)
	10%	WB	38.7	43.1	52.6	60.4	64.9	68.7	70.6	70.1	64.8	58.4	50.0	40.2	(33)
		MCDB	41.9	47.9	61.8	71.9	80.0	87.4	89.8	88.7	81.2	69.2	55.9	44.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	13.9	16.9	19.4	22.0	24.5	24.3	24.7	24.3	20.8	16.7	12.8	(35)
		MCDBR	16.6	18.6	22.5	24.8	26.6	28.0	27.1	27.0	27.3	25.3	21.8	18.0	(36)
		MCWBR	11.0	11.3	11.2	11.2	11.0	11.2	10.8	11.4	12.8	13.2	12.7	12.0	(37)
	5% WB	MCDBR	14.7	16.7	20.1	21.9	24.0	25.8	25.6	25.8	26.4	23.0	20.0	15.9	(38)
		MCWBR	10.3	10.5	10.7	10.5	10.9	11.0	11.1	11.7	13.0	12.3	11.8	11.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.347	0.397	0.467	0.495	0.492	0.494	0.483	0.448	0.433	0.466	0.417	0.350	(40)
	taud		2.222	2.085	1.937	1.915	1.977	1.982	2.020	2.113	2.089	1.938	2.030	2.203	(41)
	Ebn at Noon		250	251	246	249	253	252	253	259	253	223	217	237	(42)
	Edh at Noon		31	42	54	59	57	56	54	48	46	48	38	30	(43)
(44) All-Sky Solar Radiation	RadAvg		634	902	1282	1709	2057	2348	2372	2162	1774	1168	745	561	(44)
	RadStd		57	98	94	144	124	91	76	87	56	102	94	70	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.65	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	+207	+146	(46)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	+135	+115	(47)

Nomenclature: See separate page